THE TWO FUNCTORS ON PRESHEAVES ATTACHED TO A FUNCTOR OF CATEGORIES ARE ADJOINT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete construction of the two maps and records the functoriality of each; the sentence 'We will leave it to the reader to verify they are each others inverse' concerns the closing routine verification and is preserved verbatim inside the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The unit relating a presheaf to the composite of the two functors. Same chapter, Lemma lemma-recover. Supplies the map from a presheaf to the composite used in the first construction.
- The functors attached to a functor of categories. Same chapter, the definitions of the two functors on presheaves and of the auxiliary category of morphisms into a fixed object. Supplies the functors between which the adjunction is asserted.

Source reference. Original lemma 00VD: sources/stacks-sites/pinned-original-source.tex, source lines 805–805. Navigation pointer invoked at proof line 805; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *The functor u_p is a left adjoint to the functor u^p. In other words the formula*

$$\mathrm{Mor}_{PSh(\mathcal{C})}(\mathcal{F}, u^p\mathcal{G}) = \mathrm{Mor}_{PSh(\mathcal{D})}(u_p\mathcal{F}, \mathcal{G})$$

holds bifunctorially in $\mathcal{F}$ and $\mathcal{G}$.

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Proof. Let $\mathcal{G}$ be a presheaf on $\mathcal{D}$ and let $\mathcal{F}$ be a presheaf on $\mathcal{C}$. We will show that the displayed formula holds by constructing maps either way. We will leave it to the reader to verify they are each others inverse.

Given a map $\alpha : u_p\mathcal{F} \to \mathcal{G}$ we get $u^p\alpha : u^pu_p\mathcal{F} \to u^p\mathcal{G}$. Lemma 00VD says that there is a map $\mathcal{F} \to u^pu_p\mathcal{F}$. The composition of the two gives the desired map. (The good thing about this construction is that it is clearly functorial in everything in sight.)

Conversely, given a map $\beta : \mathcal{F} \to u^p\mathcal{G}$ we get a map $u_p\beta : u_p\mathcal{F} \to u_pu^p\mathcal{G}$. Let $V \in \mathrm{Ob}(\mathcal{D})$. We claim that the functor $u^p\mathcal{G}_V$ on $\mathcal{I}_V$ has a canonical map to the constant functor with value $\mathcal{G}(V)$. Namely, for every object (U, ϕ) of $\mathcal{I}_V$, the value of $u^p\mathcal{G}_V$ on this object is $\mathcal{G}(u(U))$ which maps to $\mathcal{G}(V)$ by $\mathcal{G}(\phi) = \phi^*$. This is a transformation of functors because $\mathcal{G}$ is a functor itself. This leads to a map $u_pu^p\mathcal{G}(V) \to \mathcal{G}(V)$. Another trivial verification shows that this is functorial in V leading to a map of presheaves $u_pu^p\mathcal{G} \to \mathcal{G}$. The composition $u_p\mathcal{F} \to u_pu^p\mathcal{G} \to \mathcal{G}$ is the desired map. $\square$